extensively to watch ration shops, banks, and religious places in every subdivision, where people gather.

Many cities like Lucknow, Raipur, Agra, Guwahati, and Meerut are using drones to ensure the effective implementation of Covid-19 lockdown and mitigate the spread of the virus in the towns.

Drone startups in India are working alongside authorities to provide drone services. FlytBase drones are delivering medicines, first aid kits, and emergency supplies to homes and hospitals, and aerial disinfection at scale. Marut Drones are now helping the Telangana government. GarudaUAV is working with the local police and administration in Noida. Navi Mumbai based ideaForge, a manufacturer of drones for defense, security, and industrial applications, has provided the equipment to Maharastra government.

The biggest challenge during this opportunity is to assemble the right kind of drone in this lockdown when spares and parts are not immediately available. Also, payload customisation for non-regular use cases is another challenge.

A drone's small size and flight capability allow it to access areas that otherwise may not be easy or possible. Typically, they are equipped with computer vision, camera, face and object recognition, and a combination of networking, robotics, and artificial intelligence (AI) to perform as demanded by the situation.

—Deepshikha Shukla

2 AERIAL REVOLUTION IN CONSTRUCTION SECTOR

Unmanned aerial vehicles (UAVs) or drones are playing an important role in the construction industry. In recent years, they have become an ideal choice for land surveying, building inspections, security surveillance, marketing, monitoring on-site activities, and so on

Unmanned aerial vehicles (UAVs) or drones have emerged as extremely viable tools for various sectors, ranging from geographic mapping and disaster management to climate change research, precision agriculture, and much more. Even while flying at lower altitudes, drones can capture aerial data with high accuracy at better speeds and lower cost when compared to manned aircraft or satellites.

In the construction industry too, drones allow easy access to difficult-to-reach sites and structures, such as unsafe steep slopes. This makes them an ideal choice for land surveying, building inspections, security surveillance, marketing, mapping data, monitoring on-site activities, logistics, demolition, and so on.

Selection of the right drone for an application results in better performance. Various parameters like camera resolution, quality of different components and the height of flight can affect the accuracy. Components of a commercial drone include flight controller, motors (preferably brushless for noise mitigation), propellers (standard/pusher), receiver and transmitter, batteries (lithium polymer), electronic speed controller (ESC), landing gear, flight controller, sensors such as light detection and ranging (LiDAR), and the like.


(Credit: www.lhsfna.org)

The flight controller is the brain of a drone where all the decisions are made. The transmitter is responsible for the transmission of the radio signals from the controller to the drone, while the receiver is responsible for their reception. These components along with sensors need to work properly in order to issue commands to the drone on time while flying. An ESC interprets signals from the flight controller, and converts them into phased electrical pulses to determine the speed of the motor. A motor with a lower rating will usually produce more torque and that with a higher rating will spin faster. A heavier propeller will require more torque from the motor for motion than a lighter one. Finding the right balance between the specifications of these components makes all the difference.

In a drone survey, aerial data is captured with RGB or multispectral cameras in cases where colour and near-infrared imaging needs to be performed simultaneously, and LIDAR payloads for measuring distances remotely. Images are taken several times from different angles based on site area and are usually saved on a memory card. They are geo-tagged with coordinates using a geo-tagging software. The images are then imported into photogrammetry software that processes this data to create elevation and 3D models, geo-referenced orthomosaics, volumetric measurements, and so on.

In one such example, a construction and asset management company, Graham construction, saved time and money by employing eBee Plus automated mapping drones to do topographic surveys for building and civil engineering projects. This is important for tracking progress in real-time without missing any corner of a site. Pix4D photogrammetry software is used to process the drone's imagery and create digital point cloud and contour model outputs.

The previously used terrestrial surveying methods were not only expensive but also risky at times, such as while working on the on-site quarries. The tedious task of data processing and volume calculation, which had to be done separately by specialists, is reduced using drones. It also saves surveyors from traversing dangerous sites, as aerial photography can be